

Memory challenges in biological computing

- Biological computers can currently store only one bit of information by altering neuron activity.
- But since neurons are living systems, their synaptic connections can change over time, even without input, leading to memory loss.
- Like humans, these systems can forget. Trained patterns may fade unless actively refreshed.
- Current work touches only on long-term memory, and that too at a very basic level.
- Short-term memory, like remembering a seven-digit number for 10 seconds, is an entirely different challenge and not yet understood in this context.
- This remains an open research area, and no stable solution exists yet.

represent the way neurons communicate with each other or respond to external signals.

To process these signals, the system uses analogue-to-digital converters (ADCs). These devices translate the analogue voltage signals from the brain into digital values that a computer can understand. When signals need to go the other way, from computer to brain, digital-to-analogue converters (DACs) convert digital instructions into precise currents sent through the electrodes.

The entire setup is controlled through a custom-built software interface. Researchers use an API, typically accessed through Python, to program the timing and type of stimulation, as well as to record and analyse the electrical activity of the neurons.

Because electrodes detect changes in ion concentration over time, they naturally pick up voltage differences. When these differences are digitised, the computer can work with them just like any other data, allowing for a direct link between digital systems and living brain tissue.

How do living neurons learn?

The question of how living neurons learn remains one of the biggest open questions in neuroscience. The short answer is: we do not really know. No one does, not entirely. But we do have some clues, and ongoing research is helping piece together parts of the process.

To begin with, it helps to understand how artificial neural networks learn. Forty years ago, we also did not know how to train artificial networks. Over time, researchers

developed methods to adjust the weights of artificial neurons, stored in matrices called tensors, allowing these systems to improve over time. This led to the powerful AI models we have today.

Living neurons have a similar goal: to change the strength of connections (called synapses) between neurons so that the output is useful. But the big difference is access and control. In artificial networks, we can read and change the weights easily. In biological systems, we cannot. Each neuron can have around 10,000 connections, and current tools only allow us to observe or adjust a very small fraction of these connections.

The basic process of changing synaptic strength is well understood and has been reproducible in laboratories for decades. However, we cannot yet modify every synapse independently in brain-like tissues. There are tools to affect one synapse, but not thousands across an entire neural structure, such as an organoid.

A very few companies are trying to train neurons in vitro (outside the body) rather than in vivo (inside a living animal). In vivo training, like teaching a rat to navigate a maze, is well studied. But in vitro training presents new challenges because the system is isolated and lacks behavioural context.

To work around this, scientists use a closed-loop approach. In this method, neurons are stimulated, and their responses are measured. Based on how close the output is to the desired result, the system adjusts its stimulation strategy. This can be done either randomly (using an

algorithm) or through a reward-based system.

A reward-based system assumes that neurons can adjust their own connections when they receive a signal that they are doing the 'right thing.' This concept, rooted in behaviourism, is similar to how animals learn with rewards like food. In laboratories, researchers sometimes release neurotransmitters like dopamine into the environment around the neurons to simulate a reward. Others are exploring more abstract ideas like the free energy principle, which relies on predictable spiking patterns being perceived as positive reinforcement.

As for how information is encoded in the brain, the best current understanding is that it is done through electrical spikes. The timing and correlation of these spikes between neurons appear to carry the information. But this may not be the whole story. Some suspect that other phenomena, possibly even involving light or unknown signals, could play a role.

Ultimately, while the fundamental mechanisms of learning in living neurons remain unclear, researchers are drawing on what we know from artificial intelligence, behaviourism, and neuroscience to push forward. It is a long-term effort, but one that could lead to major breakthroughs in both biology and computing.

Living computers represent a new and still-developing way to process information—one that uses real neurons instead of transistors. A few companies are now building the tools to interact with these systems, using electrodes, microfluidics, and software, to better understand how neurons process and learn. This approach may someday lead to new kinds of computing that can solve problems beyond the reach of digital machines. The path is complex and uncertain, but the potential makes it worth exploring. **EFY**

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